Integrity is about the truthfulness of the participants, the messages they send, the data, and the operations of the system under consideration.

> **DEFINITION** *Integrity,* in the context of blockchain, means ensuring the security and privacy of data and confidentiality of transactions.

Integrity, shown in the second quad of figure 3.1, begins with a method for uniquely identifying the peer participant on the node. In a decentralized system, no username and password identify who you are, as in a centralized system. Blockchain account address is a simple way to specify a unique identity for a participant. The elements of integrity--identity, security, privacy, and confidentiality (figure 3.1, 2a to 2d)--are based predominantly on the private-public key pair concept. You'll learn in chapter 5 how to implement security and privacy aspects for participant data, using a combination of cryptography and hashing algorithms. Likewise, you'll learn about confidentiality (figure 3.1, 2d) and its implementation in a micropayment channel application in chapter 7.

In this chapter, you'll design a smart contract for application-specific trust and integrity. The ballot smart contract illustrates verification, validation, and identity and privacy aspects. Let's explore applying these features to solve a well-known problem. Digital democracy has been a quest ever since the advent of the internet. Balloting is an exciting topic that garners the interest of a wide range of people. We'll solve a balloting problem that allows for electronic voting by a set of decentralized participants.

3.2 Digital democracy problem